Venisha Kalola

Computer

Engineering

Company: PhonePe

Job Profile: SDE

Offer Type: Internship + Placement

Internship Stipend: 80k per month

CTC: 33.5L (19.25L Base + 5L joining bonus + 8.5L Stocks)

Location: Pune / Bangalore

Process Date: - PPT + OA : 7th July, 2025

- Interview Day : 8th July, 2025

Placement Process:

1. **OA (Coding Round)** [All those who applied (200+)]

Patform : DoSelect || 90 minutes || 4 Questions

Each Question has 100 points (Total 400 points)

1. 1. There are some n friends and needed to find components, as soon as I read the question I understood it was a **DSU (Disjoint Set Union)** or **Union-Find** question so I wrote the DisJoint Class and was able to solve this question within 10-15 minutes. It was simple just had to count the total components.
   I was able to pass all test cases (12/12 testcases).

2. You are given an array of length n representing the number of laddoos distributed to n people standing in a line. Each person must receive at least **1** and at most **m** laddoos. Some people have already been assigned laddoos (represented as numbers between 1 and m), while others are yet to be assigned and are marked with a 0 in the array.
   Your task is to assign laddoos to the unassigned positions (i.e., the 0s in the array) such that the final array satisfies the condition: for every pair of adjacent people, the absolute difference between the number of laddoos they receive is less than or equal to **k**. You need to determine the **total number of valid ways** to make these assignments such that all constraints are satisfied.

   For Example : n=3, m=5, k=3.   Array = [5 3 0]
   Expected Output : 5

   Solution : So for the last person as we can assign (1 to m i.e. 5 laddoos) as the adjacent element is 3 the condition of <=k holds so here I have 5 different ways
   ([5 3 1], [5 3 2],.., [5 3 5])
   My Approach : I was able to solve it partially I used recursion , it could also be further memorized for all testcases to pass.

3. Given an Array of n elements have to find 2 elements whose sum is equal to m and the bit addition is <=k. Suppose sum should be 10 and I have elements 6 (110 in binary) and 4 (100 in binary)
   So for 6(the bit sum will be 2 + 1 = 3) and for 4 (2)
   The total sum 3+2 =5 and if it is not more than k it will be considered.

   My Approach : I had sorted the array (Time Complexity : nlogn) so that I can use 2 pointer concept to find sum = m, and if the sum condition holds then I check the bit condition .
   For checking bit condition ran loop to all bit position from 0->31
   Suppose we have 0 1 0 1 0 1 (left is msb while right is lsb)
   position : 5 4 3 2 1 0
   sum : 4 + 2 + 0 = 6 (so this way easily can just add the positions of 1)
   I was able to solve this question fully (8/8 testcases)

4. **Sorting Hat** : You are given n students and m, the maximum number of students allowed per house. There are four houses: Slytherin, Gryffindor, Hufflepuff, and Ravenclaw. Each student has a list of preferred houses they can be assigned to. Additionally, some pairs of students are friends and must be placed in different houses. Assign each student to one of their preferred houses such that no house exceeds capacity m and no friends share the same house. Determine if such an assignment is possible.
   Solution : It can be solved using backtracking as can assign in a particular friend group n try different ways and in end if all get assigned can print "Possible" else "Not Possible"
   I was able to solve it partially.

   Tip : When the number of friend relations is not specified in the input, using Scanner in Java is not ideal, as it relies on knowing how many inputs to read. In such cases, BufferedReader is preferred since it allows reading input line by line until the end. This makes it important to understand how to handle different input formats across programming languages.

In total I was able to solve 2 questions fully while other 2 partially. I had solved question 1 and then jumped to question 3 n then proceeded to solve it completely. I think had around 50 minutes to try ques 2 and 4 and was able to do that both partially. Total 30 candidates were selected for next round.

Tip : Don't waste time on 1 question first try the questions which are able to solve and even if able to solve partially some question its ok, Partial Marking does helps so try to increase the number of total testcases passed across all problems.

2. **<u>Technical Interview 1</u>** [30 students appeared this round]

This was offline interview, there were total 10 panels so the round happened in 3 batches a there were total 30 students. My Technical Round 1 happened in first batch. It was a 1 hour round where 2 Coding questions are asked.
My Interviewer didn't asked me for a Intro and told me he will be giving 2 questions and he expects to have 1 coding question with proper optimized code on paper with complexities and the

2nd problem if not able to code out completely atleast properly discussed optimized solution and having written something on paper to even stand a chance to reach to the next round.

1. Given a NxM matrix have to find the peak element

   Example : [1 2 3
   2 1 **5**
   3 1 2].

   Solution : Here Peak Element is 5 as 5 is greater than all its adjacent numbers in all 4 directions (if a number is on boundary assume out of boundary all elements have a negative infinity)

   We had to return any peak element from the matrix

   My Approach : I first explained the obvious brute force solution which would be check for each number if peak n when find peak return. It would take time complexity of O(n^2) .

   As we wanted any peak element for each row we can perform binary search space , in normal binary search we need the array to be sorted but here binary space search would work as just need any element.

   I explained with example if we have 1 4 3 5 **4** 3 2 6 1 here middle element would be 4 we see on left of 4 we have 5 which is >4 so now right boundary mid -1

   So search space : 1 4 3 5 middle = 4 which is greater than both left and right so its peak elemnt for a particular row.

   So my approach included for every row find the peak element n then just need to check for the above and below row if its peak element and can return.

   Time Complexity : O(nlogm) Space Complexity : O(1)

   Interviewer was looking for a n(logm) solution and he told my approach was correct and I proceeded to code out the whole solution on paper.

2. The second question was based on n-ary tree

   Example :

```
            1              level 0
           / \
          2   3            level 1
         /   / | \
        4   5  6  7        level 2
```

   Q queries will be given : [startnode, endnode, node]

   Example if q [2,5,7]

   If path of 2-> 5 includes nodes 2,1,3,5. From node 7 – node 3 is the nearest which is also present in path. It was a **LCA** (lowest Common Ancestor) Question and a **Competitive programming** concept **Binary Lifting** can be used for most optimal solution.

   My Approach : As the question was been explained I understood it had something to do with LCA and I told LCA and the Interviewer smiled as I was able to guess the type of question very quickly.

   For startnode and endnode first find LCA so for node 2 and 5 it would be 1 and then I told I can run a dfs for the node and find nearest node from the path, he told me I can optimize more and don't need a dfs so was able to come up with the solution where using parent node

information and the depth the traversal was not needed. I coded out the solution on paper , also implemented the LCA function.
Interviewer was satisfied , we had also discussed time and space complexity for every approach.
He asked me do I have any Competitive Programming Experience I Told yes as I had given contests on codeforces and codechef along with Leetcode. Interviewer was satisfied as I did further optimizations in the LCA Approach. It was a binary lifting question where can find using logn and its most optimized approach, interviewer was ok if not able to code binary lifting in my case as I was able to achieve a similar optimized approach and had wrote code for it.

He asked do I have any questions for him-

I asked him about NPCI and the system design for UPI payments in general, we had a discussion on NPCI and his work experience and his role in company.

3. **Technical Interview 2** [Around 14 students appeared]

My Technical Round 2 took place after the HR round.
For some students, the HR round happened before Technical Round 2. This might have been done to speed up the process or to save time.
Technical Round 2 was similar to Round 1 - we were asked to solve two coding questions on paper. I gave a quick introduction before starting the questions.

1. Given an array ["abc", "or", "xyxcba", "cba", "xz"]
   Check if a palindromic string exists after concatenation of 2 strings.
   For above example : we have abc + cba = abccba OR abc + xyxcba = abxcyxcba
   There might exist many palindromic string can return any 1.

   My Approach : First discussed the brute force where for every pair check if palindrome and then the optimized version included usage of **Trie** , in trie can store the reverse string suppose in trie abc is already stored in reverse as cba and for string cba later can see c exists and a word exists so can return , now for a case like xyxcba also forms palindrome with abc right so now see that xyx is itself a palindrome so from the rest it forms palindrome so it will form a palindrome.
   It had TC : O(n*k*k) n= length of array , k=length of a string
   I asked is there any further optimized approach and interviewer said no and I proceeded to code out the Trie solution.

2. It was the exact question from Codeforces with **Rating 2700** which is around International Grandmaster (IGM) level **-** https://codeforces.com/problemset/problem/1270/G
   My Approach : Explained the brute force and the Interviewer gave hint to focus on the constarints $i{-}n \leq ai \leq i{-}1$ and separate the dependent variables

I was able to derive that i-ai lies between (1 to n) and it was 1-indexed array so even I lied from (1 to n)
Drew edges from i to i-ai and saw it formed a cycle, I understood that the sum will be 0 where there will be cycle. He asked to explain why the cycle leads to 0.
This was a question No student in any panel was able to come to optimized solution on its own. I think this particular question was asked to see thinking process and if can reach to solution with help of hints maybe. I dry ran the approach of cycle, didn't completely wrote the code as time was up but the basic idea I wrote on paper.

Interviewer asked if I had any questions –

We had a discussion on overall experience also in PPT they had mentioned they used T-store internally in PhonePe I asked about it and had a discussion over same. Also how UPI payments can be made using SMS service too.

3. **Hiring Manager Round** [Around 11 students appeared]

During the interview, I began by greeting the interviewer with a "Good Afternoon." He asked me to start with my internship experiences, where I explained both roles, my responsibilities, and key learnings. We then moved on to the projects on my resume - two web projects and one mobile application. I had the app on my phone and offered to demonstrate it, which he agreed to. The app included an ML model, leading to a discussion on how the model was integrated and called from the backend. He appreciated the design of the app. He then asked if I had worked on any team projects. I shared a project that contributed to my team winning an international hackathon featuring participants from over 10 countries.

Since I had mentioned teaching experience on my resume, he asked how I got that opportunity and what I taught. I explained that I had taught 11th and 12th-grade students, helping them with computer science and clearing doubts in other subjects for college entrance exams. I also spoke about my volunteering experience with an NGO, where I taught 7th and 8th graders. The NGO had on learning about my prior teaching background asked me to help students prepare for Olympiad topics. Following that, we discussed one of my major projects in detail, including the system architecture. I explained the flow using paper diagrams and covered topics like authentication, how OAuth works, OTP-based verification, and how API calls are made.

Toward the end, the interviewer asked about my most-used application, and I said YouTube. He then asked me to design a system for YouTube Shorts. I drew a high-level system design on paper and explained various components, including how the system dynamically adapts video quality based on network bandwidth. We also discussed why I am pursuing a minor in management from SP Jain, and wrapped up with a few HR-type questions - one of which was how I handled conflict. I shared an example from the JPMC Code for Good Hackathon, explaining how our team resolved a disagreement.

This round primarily tested knowledge of my resume and understanding of system design.

When Interviewer asked do I have any questions – I asked 1-2 questions and discussed topics related to it. We also had discussion over if in future payments can be done using facial recognition and the challenges and how it can be misused.

A total of 7 students received the offer, and I was one of them.

Useful Tips:

1. For OA prepare all the topics. Should know all the common patterns and able to solve LeetCode medium- hard problems.
2. Can follow Striver's series but I also recommend to do blind 75 questions https://leetcode.com/studyplan/leetcode-75/ - for clearing up basics before proceeding to the advance topics.
3. Stay calm and confident during Interview. Think out loud during the technical interviews. Even while coding it out keep on explaining and discussing the approach.
4. Interviewers are helpful in case if not able to proceed they can give hints, after hints try to think on them as they give idea about how to proceed to a particular problem.
5. Doing **CP (Competitive Programming)** is highly recommended as this year CP topics were asked. Participating in contests on platforms like Codeforces can significantly enhance your problem-solving skills and help you get accustomed to the pressure of live contests.

All the Best for your placements

Feel free to reach out for any doubts and guidance,
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